

MATH SURVEY



STUYVESANT H.S.

Spring 1986

We dedicate this magazine to the members of the Senior Math Team of the Class of 1989. May they always remember that $1989 = 17 \times 117$.

Math Survey is the official organ of the Mathematics Department of Stuyvesant High School. It represents an attempt to communicate the beauty and joy of mathematics to the high school community at an elementary level.

It can be noted that buildings from several major cities appear in the skyline depicted on the cover. This is meant to indicate our pride in Math Survey's status as one of the few high school publications to enjoy a truly nationwide circulation.

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Errata

Since publication, a number of errors have been brought to our attention. Please take note of the following corrections:

- In the note "Hamiltonian Graphs" by Alexander Ng, a definition is stated incorrectly. At the beginning of the second paragraph on page 9, the definition of the set T should read, "If a vertex v_j is adjacent to v_L , then v_{j+1} is in T ." In addition, the last sentence but one of this article should begin, "However, S and T are disjoint subsets of $\{v_1, v_2, \dots, v_L\}$."
- On page 12, in the Regents Refresher section, two comments under the heading "Equations of Plane Curves" are erroneous. After the equations of the ellipse and the hyperbola, the phrase "where a and b are the lengths of the major and minor semi-axes" should appear. At the top of page 13, the equation " $\cot \theta = x/y = 1/\cot \theta$ " appears. This is incorrect; the correct equation is " $\cot \theta = x/y = 1/\tan \theta$."
- The answers to the sample math team problems appearing at the bottom of page 17 are as follows:
 - (1) 10^{147}
 - (2) 16
 - (3) 302330880000000
 - (4) $11/16$
 - (5) $\sqrt{3}$
 - (6) No.
 - (7) $n = 333$; $a_1 = a_2 = a_3 = \dots = a_{332} = 3$; $a_{333} = 4$.
- In the article "Prime Number Distribution," in the centre of page 19, the expression " $m = \pi(n) \leq 1 + \log_2(\log_2 n)$ " is in error. The inequality should be reversed, so that it reads " $\pi(n) \geq 1 + \log_2(\log_2 n)$."
- The coupons in the advertisement appearing on page 18 were inadvertently marked with expiration dates which render them currently invalid. These coupons will in fact be honored by the dealer until May 17, 1989; provided the redeemer presents them with this sheet.

Math Survey

Volume LX

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The Mathematics Publication of Stuyvesant High School

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Special thanks to Mr. Clancy, Mr. Geller, and Mr. Greenfield for their timely assistance.

A Brief Note on Mathematical Induction

by Andrew W. Chen

Consider the statement $1+2+3+\cdots+n = \frac{n(n+1)}{2}$.

Is it true or false? Clearly it is not a statement at all, since it is impossible to say whether it is true or false, or even meaningful. When $n=7$, this "statement" is true; when $n = \text{Darien's nose}$, the "statement" becomes meaningless. It seems that its truth value is dependent on the value of n .

Suppose we modify the statement to read

$$\text{When } n=1, 1+2+3+\cdots+n = \frac{n(n+1)}{2}$$

This "statement" is now a statement, one which happens to be true. How would you prove it? You could substitute 1 for n in the equation, yielding

$$1 = 1(2)/2, \quad \text{a true assertion.}$$

A similar statement,

$$\text{When } n \text{ is an element of } \{1,2\}, 1+2+3+\cdots+n = \frac{n(n+1)}{2},$$

can be proven by substituting 1 and 2 into the equation and noting that it is true in each case:

$$1 = 1(2)/2, \text{ and} \\ 1+2 = 2(3)/2.$$

In fact, this method will work for any finite set of values for n .

What about the statement,

$$\text{When } n \text{ is an element of the set } N, 1+2+3+\cdots+n = \frac{n(n+1)}{2},$$

where N is the set of natural numbers? How would you prove this statement? To be sure, you could use the same method you used for finite sets, and substitute for every element in the set. There is, however, another method; one that will take a finite amount of time. This technique is called the Principle of Mathematical Induction.

The Principle of Mathematical Induction is based on the following type of reasoning:

Take these two statements:

$$2 + 2 = 4. \\ \text{If } 2 + 2 = 4, \text{ then grass is green.}$$

Had you proved these two statements, what could you conclude? It is considered valid reasoning to conclude that grass is green.

Let us replace " $2 + 2 = 4$ " with " $\text{When } n = 1, 1+2+3+\cdots+n = n(n+1)/2$," a statement we have proved.

Let us further replace " $\text{If } 2 + 2 = 4, \text{ then grass is green}$ " with " $\text{If, when } n = 1, 1+2+3+\cdots+n = n(n+1)/2, \text{ then, when } n=2, 1+2+3+\cdots+n = n(n+1)/2$ ", which can be proved as follows:

$$\text{Assume that } 1 = 1(2)/2; \\ \text{then } 1 + 2 = 1(2)/2 + 2$$

$$\begin{aligned}
 &= 1(2)/2 + 2(2)/2 \\
 &= 3(2)/2 \\
 &= 2(3)/2
 \end{aligned}$$

We can now conclude that when $n=2$, $1+2+3+\cdots+n = n(n+1)/2$.

For convenience, let $P(k)$ represent the statement

“When $n = k$, $1+2+3+\cdots+n = n(n+1)/2$ ”.

For example, $P(17)$ is short for “When $n=17$, $1+2+3+\cdots+n = \frac{n(n+1)}{2}$ ”

We have now shown that

$P(1)$
and If $P(1)$, then $P(2)$
are both true statements. We can thus conclude $P(2)$.

To prove $P(3)$ we would have to show

If $P(2)$ then $P(3)$;
we could then prove $P(4)$ by proving
If $P(3)$, then $P(4)$.

Note that each of these statements is of the form

(*) If $P(k)$, then $P(k+1)$
and that in order to prove $P(j+1)$, we could prove these statements for all k in the set $\{1, 2, 3, \dots, j\}$. If we were to prove (*) for every natural number k , we could conclude:
For every natural number k , $P(k)$.

This principle ought to seem reasonable to you, and is often postulated in the definition of the natural numbers. This is the Principle of Mathematical Induction.

Let us demonstrate this technique, using

$P(k) = \text{“when } n = k, 1+2+3+\cdots+n = n(n+1)/2\text{.”}$

First we prove $P(1)$.

$1=1(2)/2$, as desired.

Now, we must prove for each natural number k ,

If $P(k)$, then $P(k+1)$.

Assume $P(k)$; that is,

When $n=k$, $1+2+3+\cdots+n = \frac{n(n+1)}{2}$,

or $1+2+3+\cdots+k = \frac{k(k+1)}{2}$;

$$\begin{aligned}
 \text{then } 1+2+3+\cdots+k+(k+1) &= \frac{k(k+1)}{2} + (k+1) \\
 &= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} \\
 &= \frac{(k+2)(k+1)}{2} \\
 &= \frac{(k+1)(k+2)}{2} \\
 &= \frac{(k+1)[(k+1)+1]}{2}
 \end{aligned}$$

Therefore when $n = k+1$, $1+2+3+\cdots+n = \frac{n(n+1)}{2}$, as advertised.

We conclude that, for every natural number k , when $n = k$,

$$1+2+3+\cdots+n = \frac{n(n+1)}{2}. \quad \text{Q.E.D.}$$

(continued on page 9)

Recent Advances in Linear Programming

by Satip Ghosh

Narenadra Karmarkar, a 29-year-old mathematician at Bell Labs in Murray Hill, New Jersey, has made a remarkable breakthrough in solving optimization problems that often prove to be too vast or complex for even the most powerful computers. This discovery has set off inquiries from many elements of the private sector, ranging from oil companies to airlines, who have big money at stake in linear programming problems. These problems involve highly complicated systems, often with thousands of variables; they arise in contexts such as the allocation of time on communications satellites. AT&T believes that Karmarkar's method will provide a new approach to the problem of routing telephone calls through cities with maximum efficiency. Because of the different volumes of calls between cities, the different capacities of telephone lines, and the different needs of users at various hours, the problem is extraordinarily difficult. Linear programming is particularly useful whenever a limited, expensive resource must be spread most efficiently among many users. Dr. Karmarkar has devised a new procedure that speeds up the handling of such problems. To solve these problems, computers use algorithms which involve examining a set of tentative solutions in order to obtain the one which minimizes or maximizes some quantity. An efficient algorithm is thus one which examines relatively few answers before obtaining the best one.

Dr. Narendra Karmarkar, whose father and uncle were both excellent mathematicians, was born in Givalior, India. He grew up in Poona, near Bombay, and received his education in India. He joined Bell Labs in 1983 after attending the California Institute of Technology and acquiring his doctorate from the University of California at Berkeley.

Linear programming problems are usually visualized as geometric solids with many faces. Each corner of each face represents a possible solution. The task of the algorithm is to find the best solution, say the corner at the top, without having to calculate the location of each one. The simplex method, devised by the mathematician George B. Dantzig in 1947, runs along the edges of the solid, checking one corner after another but always heading in the direction of the best solution. In practice, it manages to get there efficiently enough to solve some problems (in which the number of variables is between 15,000 and 20,000). When implemented on a high-speed mainframe computer, the most effective version of the simplex algorithm can locate the best choice in a situation involving 10^{1000} options (the size of a typical linear programming problem) in a few hours.

The Karmarkar algorithm, by comparison, takes a giant shortcut. In terms of the earlier analogy, it plunges through the middle of the solid. After selecting an arbitrary interior point, the algorithm warps the entire structure — in essence, reshaping the problem's geometric form — in a way designed to bring the chosen point exactly into the center. The next step is to find a new point in the direction of the best solution and to warp the structure again, bringing the new point into the center. Dr. Karmarkar comments: "Unless you do this warping, the direction that appears to give the best improvement each time is an illusion." The repeated transformations, based on a technique known as projective geometry, lead rapidly to the best answer. The Karmarkar algorithm, unlike previous methods, can sort through 10^{100000} possible options. Dr. Karmarkar estimates that, "If the simplex method requires n steps to reach optimality, my method requires only the logarithmic function of n ." A computer, using his algorithm, would take a few minutes to solve a typical linear programming problem at a cost of a few dollars.

AT&T is currently coding Dr. Karmarkar's algorithm, since the algorithm itself cannot be patented or copyrighted. Dr. Karmarkar has published his paper formally in the journal Combinatorica.

Dr. Karmarkar's algorithm has been received with great enthusiasm on all fronts. Attendance at a recent mathematics conference increased 30% because of Dr. Karmarkar's presence. Many regard his algorithm as a major breakthrough. Work is currently being conducted on the numerous practical applications of Dr. Karmarkar's method.

Hamiltonian Graphs

by Alexander Ng

Take a look at Figure 1. As we can see, it is a geometric figure consisting of vertices and interconnecting edges. Such a figure is called a graph. For some reason, mathematicians actually study the properties of these figures—it's called graph theory.

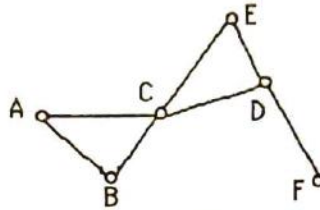


figure 1

So let's talk about graph theory. First, we need some definitions:

(i) The degree of a vertex x (denoted by $\deg x$) is defined as the number of edges protruding from the vertex. Thus, in the graph above, vertex A has degree 2, while vertex C has degree 4.

(ii) Two vertices are adjacent if they are joined by an edge. For example, A and B are adjacent, while A and F aren't.

To the graph theorist, the manner in which the graph is drawn is irrelevant; he concerns himself only with the interconnections between vertices and edges. Indeed, he considers himself to be completely informed of the structure of a graph once he knows which pairs of vertices are adjacent.

An interesting type of graph is the Hamiltonian graph. These graphs are so named because of a toy puzzle invented by William Rowan Hamilton in 1859. Hamilton's puzzle was a wooden dodecahedron (a twelve-faced polyhedron). The solid represented the Earth; the edges were roads, and the vertices symbolized the cities of the world. The object of the puzzle was to start at a particular city, then visit every other city of the dodecahedron precisely once, without going back over previously traversed edges, and finally return to your original city. In other words, find a globe-spanning round trip.

Since Hamilton's puzzle is essentially a graph, (Think about it! It's made up of vertices and edges.), we can generalize the idea behind his toy puzzle.

Suppose a graph G has N vertices. Then G is Hamiltonian if there exists a sequence of (distinct) vertices $v_1, v_2, v_3, \dots, v_N, v_1$ where each vertex is adjacent its successor. This sequence of vertices is called a Hamiltonian cycle.

At first reading, this definition may seem unfathomable. But all I am saying is that G is Hamiltonian if you can find a path starting at some initial vertex which travels over every vertex just once and then returns to the initial vertex. Let us look at some examples of these graphs.

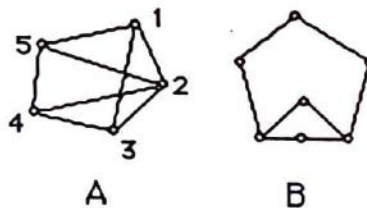


figure 2

It is obvious that graph A has a Hamiltonian cycle. Starting at vertex 1, go to 2, 3, 4, 5, and then return to 1. Note that a graph may have more than one Hamiltonian cycle; graph A has

others, such as 1, 2, 5, 4, 3, 1. Are there more? In contrast, B has no Hamiltonian cycle. (Try looking for one. You will not succeed.)

One of the major unsolved problems of graph theory is finding some quick and easy way of determining whether a given graph is Hamiltonian. Although there is no known necessary and sufficient condition to tell if a graph is Hamiltonian, some progress has been made. Intuitively, mathematicians say that if a graph has lots and lots of edges then chances are quite good that it is Hamiltonian. Similarly, if the degree of every vertex is huge, then, again, the graph will probably contain a Hamiltonian cycle. The theorem of Ore, quoted below, provides a sufficient condition for a graph to be Hamiltonian: Some graphs are Hamiltonian and yet do not satisfy the hypothesis of Ore's Theorem. The theorem is thus not particularly useful, but we'll prove it anyway.

Theorem (Ore): If G is a connected graph with $n \geq 3$ vertices, with the property that, for any 2 non-adjacent vertices x and y , we have

$$\deg x + \deg y \geq n,$$

then G is Hamiltonian.

proof: Let us assume that G is not Hamiltonian; we will then arrive at a contradiction in order to prove the theorem. We begin with a definition. A path is a sequence of distinct vertices $v_0, v_1, v_2, \dots, v_L$, where each vertex in the sequence after v_0 is adjacent to its predecessor. In particular, we say that this path has length L . Clearly G has some path (not necessarily unique) of maximum possible length; we depict this maximal path in figure 3 below.



figure 3

Note that $v_0, v_1, \dots, v_L$ may be all the vertices in G —and then again, they may not. In either case we shall prove that there is no cycle of length $(L+1)$ in G . We justify this proposition as follows: In the first case, if the maximal path does contain every vertex in G , then a cycle of length $(L+1)$ would make G Hamiltonian, contrary to our assumption. In the latter case, suppose that $n > L+1$; that is, that G has some vertices not lying on the maximal path. Then, if there were a cycle of length $(L+1)$, we would have a configuration like figure 4a. Since G is connected and has some other vertices not on the cycle, we know there is at least one edge joining a vertex on the cycle to one not on the cycle, as in figure 4b. The contradiction is immediate since there would then be a path of length $(L+1)$ in G , which contradicts the maximality of L . (See figure 4c.)

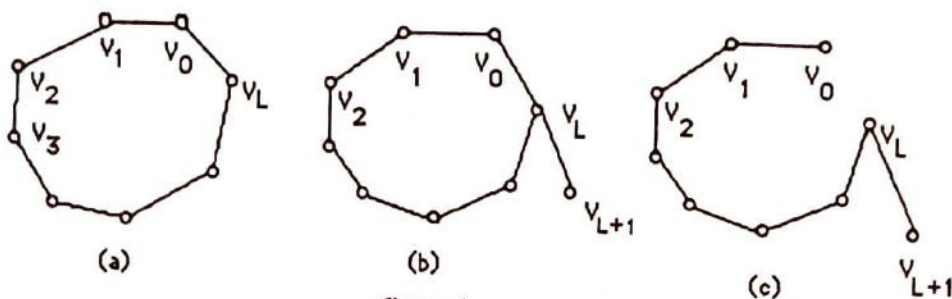


figure 4

Now let's return to the path in figure 3. We claim that the following scenario cannot occur: v_0 is adjacent to some vertex v_j on the path while v_L is adjacent to its predecessor v_{j-1} . The reasoning is simple: If this did occur, then we would obtain figure 5. It is evident in 5b that G will contain a cycle of length $(L+1)$, but we have proven above that this cannot occur.

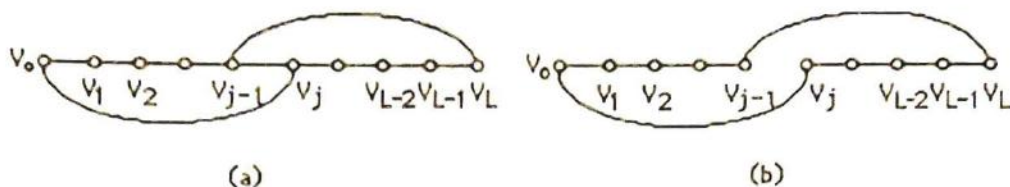


figure 5

Let S be the set of vertices that are adjacent to v_0 . Note that S is a subset of the vertices on the path. This holds true because if v_0 were adjacent to some vertex u not on the path then there would be an even longer path $u-v_0-v_1-\dots-v_L$. Likewise, v_L avoids every vertex off the path.

Now let T be the set of vertices on the path defined as follows:

If a vertex v_j is adjacent to v_L , then v_{j-1} is in T . (Why have we constructed these two sets? Because they will lead to the desired contradiction!)

We observe two interesting properties of S and T . First, S and T are disjoint, for if they had a common element, then we should obtain figure 5, which as noted above fails to occur. Again, the number of elements of S is equal to the degree of v_0 , while the number of vertices in T is the degree of v_L . Thus, letting $|A|$ denote the number of elements in the set A , we have

$$|S| + |T| = \deg v_0 + \deg v_L \geq n, \text{ by hypothesis.}$$

However, S and T are two disjoint subsets of $\{v_0, v_1, v_2, \dots, v_{L-1}\}$, and so must satisfy

$$|S| + |T| \leq L < n.$$

This contradicts the previous inequality, and the theorem is proved.

Induction

(continued from page 5)

The Principle of Mathematical Induction is a very important and powerful technique, useful not only in algebra but also in geometry, graph theory, set theory, number theory, and the solution of such curious problems as the Tower of Hanoi. It can be shown to be equivalent to the Principle of the Least Natural Number, which says that any non-empty set of natural numbers contains a least element. The Principle of Mathematical Induction can be applied to any set whose elements can be put into an order which satisfies a corresponding Principle of the Least Element. The wide application of mathematical induction follows from a principle of set theory known as the Axiom of Well-Ordering, which asserts that every set can be so ordered. This axiom, however, often leads to seeming paradoxes and is therefore suspect. The Principle of Mathematical Induction is thus seen to lie at the heart of a fundamental controversy in the theory of sets.

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Computer Bitblt Rotation

by Thomas Witelski

Computers display information to us via a video display screen. The display is basically a rectangular arrangement of points. Each point is stored by the computer as one or more bits of memory. Bitblt, bit-block transfer, operations manipulate rectangular blocks of bits and therefore change rectangular areas of the screen.

The simplest bitblt operation copies a block from one position to another on the screen:

```

a b c . . . . . a b c . . . . .
d e f . . . . . d e f . . . . .
. . . . . --> . . . . .
. . . . . . . . a b c . .
. . . . . . . . d e f . .

```

In Macintosh Pascal, this bitblt command is called CopyBits. Copying a block will not erase the original block. It is more efficient to copy a large block than many individual points because the computer operating system is better suited to handling fewer, even if more lengthy, commands.

Each bitblt operation treats the block that it works on as a single entity; it cannot break up or change the block. There is no special bitblt command that rotates an image. Computers are not well suited to rotating images because the calculations involved often require approximations for the irrational values of trigonometric functions. Rotation by 90 degrees is a simpler matter: It requires only that the computer copy each point of the image from its coordinates (x,y) to its new coordinates (-y,x). If the image is complex, however, it may be inefficient to copy all of it point by point. There is a faster way using bitblt commands.

By using bitblt copy operations in my method it is possible to rotate a square image (or one that could be put in a square outline) n by n points in size in 3n-1 steps, rather than n² steps. All that is required is a workspace of 2n-1 by 2n-1 points. In the following example n=3.

```

a b c . .      a b c . .
d e f . .      d d e f .
g h i . . --> g h g h i
. . . . .      . . . . .
. . . . .      . . . . .

```

Each row of n points after the first one is shifted to the right by its row number minus one points.

```

a b c . .      a b c . .
d d e f .      d d e f .
g h g h i --> g h g h i
. . . . .      . . . a b
. . . . .      . . . d d

```

The top leftmost square block that is n-1 by n-1 is copied to a position n points right and n points down.

a b c . .	a b c f i
d d e f .	d d e h b
g h g h i	g h g a d
. . . a b	. . . a b
. . . d d	. . . d d

-->

Each vertical column of n points whose column number is $n+1$ to $2n-1$ is shifted up by its column number minus n points.

a b c f i	a b c f i
d d e h b	h b e h b
g h g a d	a d g a d
. . . a b	. . . a b
. . . d d	. . . d d

-->

The block of $n-1$ by $n-1$ points at the second row and the $n+1$ th column is copied to the 2nd row, the 1st column.

a b c f i	c f i f i
h b e h b	b e h h b
a d g a d	a d g a d
. . . a b	. . . a b
. . . d d	. . . d d

-->

Each row of n points from the 1st to the $n-1$ th is shifted to the left by n minus its row number of points.

A B C	C F I
D E F	B E H
G H I	A D G

-->

Therefore the image has been rotated by 90 degrees (in this case counterclockwise) in $(n-1)+1+(n-1)+1+(n-1)=3n-1$ steps. This is the minimum number of steps required to rotate an n by n square image by 90 degrees.

If a three dimensional object was cut into two dimensional slices and these 2-d images were stored in a computer, the object would in effect be stored. The stored object could then be rotated about its three axes (x, y , and z) by using modified versions of the two dimensional bitblt rotation. A cube could be rotated 90 degrees about the z axis by rotating it about the x axis, then about the y axis, then, in the opposite direction about the x axis. This would result in a net z rotation, but it would not be the minimum number of steps in which such a rotation could be done. The plane in which the data slices are given is the rotation which requires the greatest number of steps.

Bitblt 90 degree rotation works not only for 2-d and 3-d but also for higher-dimensional structures. Many charts could be made in terms of two variables (age and height, for example) which together would describe the one four dimensional structure. From one set of charts, all in the same two coordinates, to produce a set of charts in a different set of two coordinates would require a rotation of the 4-d structure. In general, for an object of d dimensions there would be $d(d-1)/2$ ways to rotate the object. Each of the rotations could be performed by a series of bitblt operations working on a plane representation of the object.

Bitblt rotation has applications in making computer-related work faster. For example, the use of bitblt would make CAT scanners, CAD/CAM's, and databases work faster.

Regents Refresher

Due to the New York State Board of Regents' introduction of a new high school mathematics curriculum, it is more difficult for us to provide overview of the material covered than in previous years. Readers who are in the classic courses—algebra, geometry, and trigonometry, designated M1/2, M3/4, and M51/52—should have no difficulty in recognising the material required for their exams. For the benefit those in sequential math—MQ1/2, MQ3/4, and MQ5/6—we have prefixed the appropriate sections with roman numerals, thus indicating the year in which the material is taught.

This section does not pretend to be a complete review. It includes only the most important theorems and formulæ.

(II,III) Equations of Plane Curves:

Line: $y=mx+b$ where m = slope and b = y-intercept.

$y-y_1 = m(x-x_1)$ where (x_1, y_1) is a point on the line and m is the slope.

Slopes of parallel lines are equal.

The product of the slopes of two perpendicular lines is -1 .

Circle: $(x-h)^2 + (y-k)^2 = r^2$, where (h,k) is the center of the circle with radius, r .

Ellipse: $x^2/a^2 + y^2/b^2 = 1$, where a and b are the major and minor axes.

Parabola: $(x-h)^2 = 2p(y-k)$, where (h,k) is the vertex and p is the distance from the vertex to the directrix

or $y = ax^2 + bx + c$, where $x = -b/2a$ is the axis of symmetry.

Hyperbola: $x^2/a^2 - y^2/b^2 = 1$, where a and b are the major and minor axes.

(III) Logarithms: " $\log_b N = E$ " means $b^E = N$.

$$\log_c 1 = 0$$

$$\log_c c = 1$$

$$\log_c c^b = b$$

$$\log_c a^b = b \log_c a$$

$$\log_c ab = \log_c a + \log_c b$$

$$\log_c 1/a = -\log_c a$$

$$\log_c (a/b) = \log_c a - \log_c b$$

$$\log_c \sqrt[b]{a} = (\log_c a)/b$$

$$\log_b a = \log_c a / \log_c b = \log_b c / \log_a c$$

(II) Laws of Exponents:

$$x^m \cdot x^n = x^{m+n} \quad (x^m)^n = x^{mn} \quad x^m / x^n = x^{m-n} \quad x^{-m} = 1/x^m \quad x^0 = 1$$

$$x^{1/m} = \sqrt[m]{x} \quad x^{m/n} = \sqrt[n]{x^m} = (\sqrt[n]{x})^m$$

(II) The Quadratic Formula:

The roots of the equation $ax^2 + bx + c$ are

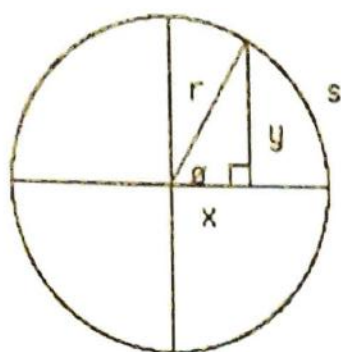
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

If the discriminant, $b^2 - 4ac$, is positive: the roots are real and distinct.

zero: the roots are real and equal.

negative: the roots are complex conjugates.

(III) Trigonometry:



SOH-CAH-TOA

$$\sin \theta = y/r$$

$$\cos \theta = x/r$$

$$\tan \theta = y/x$$

$$\csc \theta = r/y = 1/\sin \theta$$

$$\sec \theta = r/x = 1/\cos \theta$$

$$\cot \theta = x/y = 1/\tan \theta$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$\cot^2 \theta + 1 = \csc^2 \theta$$

$$\sin \theta = \cos(90^\circ - \theta)$$

$$\tan \theta = \cot(90^\circ - \theta)$$

$$\sec \theta = \csc(90^\circ - \theta)$$

$$s = r\theta$$

$$r^2 = x^2 + y^2$$

Sum/Difference Formulas: $\sin(x \pm y) = \sin x \cos y \pm \sin y \cos x$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

Double Angle Formulas: $\sin(2x) = 2\sin x \cos x$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

$$= 2\cos^2 x - 1 = 1 - 2\sin^2 x$$

$$\tan(2x) = \frac{2\tan x}{1 - \tan^2 x}$$

(Note: the double angle formulas can be derived from the sum formulas by letting $x=y$.)

Half Angle Formulas: $\sin^2(x/2) = \frac{1 - \cos x}{2}$ $\cos^2(x/2) = \frac{1 + \cos x}{2}$

$$\tan^2(x/2) = \frac{1 - \cos x}{1 + \cos x}$$

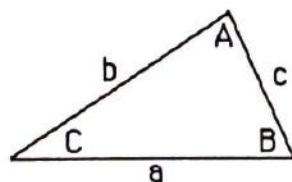
In the graph of $y = N \sin rx$, the amplitude is N , and the period is $2\pi/r$.

360 degrees = 2π radians.

To convert from degrees to radians, multiply by $\pi/180$.

To convert from radians to degrees, multiply by $180/\pi$.

Triangles:



$$\text{Area} = (1/2)ab\sin C$$

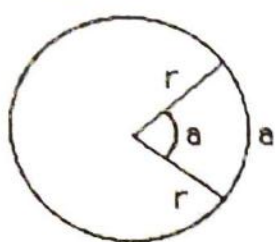
$$= (1/4)\sqrt{(a+b+c)(a+b-c)(b+c-a)(c+a-b)}$$

$$\text{Law of Sines: } \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \text{diameter of circumcircle}$$

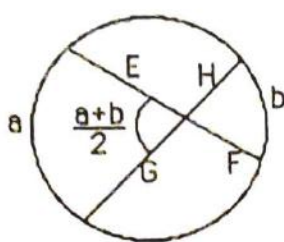
$$\text{Law of Cosines: } c^2 = a^2 + b^2 - 2ab\cos C$$

$$\text{Law of Tangents: } \frac{\tan((A+B)/2)}{\tan((A-B)/2)} = \frac{a+b}{a-b}$$

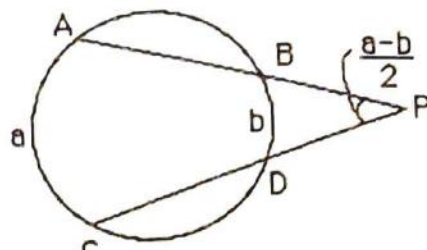
(III) Geometry:



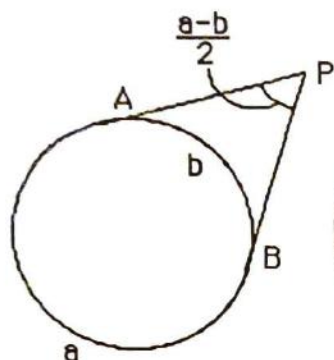
$r = \text{radius}$



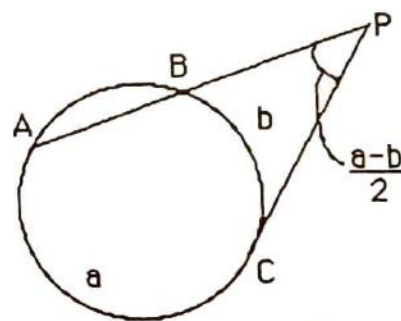
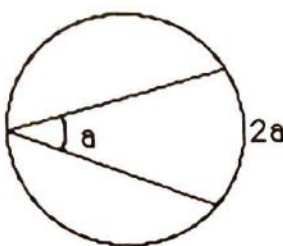
$EF = GH$



$(PA)(PB) = (PC)(PD)$



$PA = PB$



$(PA)(PB) = (PC)^2$

The required proofs for the Geometry Regents may be found in the Dressler text, pp. 733-737. The following are the numbers of the required proofs; 3,5,6,7,8,9,10,11,12,13.

(II) (A=Angle, S=Side, b=base, h=altitude)

Two triangles are congruent if you can prove SSS, SAA, SAS, or ASA.

Two triangles are similar if you can prove AA.

Two right triangles are congruent if you can prove their hypotenuses are congruent and one pair of legs are congruent. (HL)

The sum of the angles in a triangle is 180° .

The base angles of an isosceles triangle are congruent.

Area of triangle $= (\frac{1}{2})bh$.

Area of a trapezoid $= (\frac{1}{2})(b_1 + b_2)h$.

Area of a parallelogram $= bh$.

Area of a regular n-gon $= (\frac{1}{2})ap$ (where a=apothem, p=perimeter).

In a right triangle with legs a and b and hypotenuse c, $a^2 + b^2 = c^2$.

In a circle, the diameter that is perpendicular to a chord bisects the chord.

The locus of points at distance r from point P is the circle with center P and radius r.

The locus of points equidistant from two points P and Q is the perpendicular bisector of PQ.

The locus of points equidistant from two lines l and m are the bisectors of the angles formed by l and m.

(B=area of base, h=height, r=radius)

Volume of a prism $= Bh$

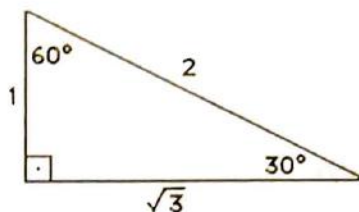
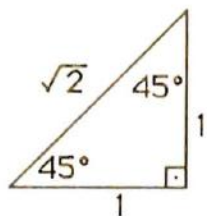
Volume of a cylinder $= \pi r^2 h$

Volume of a pyramid $= (\frac{1}{3}) Bh$

Volume of a cone $= (\frac{1}{3}) \pi r^2 h$

Volume of a sphere $= (\frac{4}{3}) \pi r^3$

The required constructions for the Geometry Regents may be found in the Dressler text, pp. 616-639. The following are the numbers of the required constructions; 5-7, 12, 13, 18-22, 25, 27-29.



(I, III) Probability:

Let A and B represent events in a sample space.

If A and B are independent, then $P(A \cap B) = P(A) \cdot P(B)$.

If A and B are disjoint, then $P(A \cap B) = 0$ and $P(A \cup B) = P(A) + P(B)$.

If the probabilities are uniformly distributed, event A can occur in $n(A)$ ways, and there are $n(S)$ possible outcomes in S, then $P(A) = n(A)/n(S)$.

The number of permutations of n objects is $n! = n(n-1) \cdot \dots \cdot (2)(1)$.

The number of permutations of n objects, k of which are indistinguishable, is $n!/k!$.

The number of permutations of r objects chosen from a set of n objects is

$${}_nP_r = n!/(n-r)! = n(n-1)(n-2) \cdots (n-r+1)$$

The number of combinations of r objects chosen from a set of n objects is

$${}_nC_r = n!/r!(n-r)! = {}_nP_r/r!$$

(III) The binomial theorem:

$$(a+b)^n = \sum_{r=0}^n {}_nC_r a^r b^{(n-r)} = a^n + {}_nC_1 a^{n-1} b + {}_nC_2 a^{n-2} b^2 + \cdots + {}_nC_{n-1} a^1 b^{n-1} + b^n$$

(I,II) Logic:

Laws of Inference

Symbolic Forms

1. Law of Detachment
(Modus Ponens)

$$[(p \rightarrow q) \wedge p] \rightarrow q$$

2. Law of the Contrapositive

$$(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$$

3. Law of Modus Tollens

$$[(p \rightarrow q) \wedge \sim q] \rightarrow \sim p$$

4. Chain Rule (Law of Syllogism)

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$$

5. Law of Disjunctive Inference

$$[(p \vee q) \wedge \sim p] \rightarrow q$$

$$[(p \vee q) \wedge \sim q] \rightarrow p$$

6. Law of Double Negation

$$\sim(\sim p) \leftrightarrow p$$

7. De Morgan's Laws

$$\sim(p \wedge q) \leftrightarrow (\sim p \vee \sim q)$$

$$\sim(p \vee q) \leftrightarrow (\sim p \wedge \sim q)$$

8. Law of Simplification

$$(p \wedge q) \rightarrow p$$

$$(p \wedge q) \rightarrow q$$

9. Law of Conjunction

$$(p) \wedge (q) \rightarrow (p \wedge q)$$

10. Law of Disjunctive Addition

$$p \rightarrow (p \vee q)$$

(I, III) Statistics

Measures of Central Tendency:

The mean, or arithmetic mean, of a set of n numbers is the sum of the numbers divided by

n . Therefore, for a sample of scores, $x_1, x_2, x_3, \dots, x_n$:

$$\text{The mean} = \bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

When a set of data is arranged in numerical order, the "middle" score is called the median. If there is an even number of terms, then the median is found by averaging the two "middle" terms.

The mode is the score that appears most often in a set of data. For some sets of data there can be more than one mode (when two modes appear, the data is called bimodal) and, in some cases, no mode whatsoever.

Measures of Dispersion:

The range of a set of data is the difference between the highest and the lowest scores.

The mean absolute deviation of a set of data is the average value of the difference between the data and the mean. In symbols:

$$\text{mean deviation} = (1/n) \sum_{i=1}^n |x_i - \bar{x}|$$

The variance, v , of a set of data is the average of the squares of the deviations from the mean. In symbols:

$$\text{variance} = v = (1/n) \sum_{i=1}^n (x_i - \bar{x})^2$$

The standard deviation, s , of a set of data is equal to the square root of the variance. In symbols:

$$\text{standard deviation} = s = \sqrt{v}$$

Math Team

by Bill Ping Yang

The 1986 spring math team season has finally come to an end. The season, which has been marked by injury and scandal, terminated in April.

The once-proud senior team was shaken by the loss of its most illustrious members. Captain Darien G. Lefkowitz was barred from competition during the months of February and March for illegal use of an abacus; Captain Bill Rugolsky was disqualified in early March on moral grounds for impersonating another student; Captain Louis Tao was removed from competition after developing a severe paper cut. Captain Tony Friedman withdrew of his own volition, citing religious scruples. His final words were, "I hate \$\$\$*ing Wednesdays!" (The full impact of the wave of Holistic Abominism sweeping through mathematical competitors has yet to be assessed.) The remaining captain, Peter Ying, might have made it through the season had he not been disqualified in early April after being implicated in the AIME point-shaving scandal. The only comment he offered to reporters was, "Well, at least I pay my parking tickets."

Somehow, throughout the scandal and tragedy, the senior team managed to prevail (possibly due to the abundance of tall, intelligent freshmen with false moustaches). The team attained a perfect score of 180 in the statewide New York Mathematics League competition and a winning 330 out of a possible 360 in its first year of competition in the Atlantic-Pacific High School Mathematics League. Also in this competition, Mr. Lefkowitz contrived to retain a perfect score. A special committee is being appointed to investigate the matter.

It is rumoured that Stuyvesant has maintained its ascendancy in the event of the year, the American High School Mathematics Examination. Our team score of 400 out of a possible 450 is known to be the highest in the city, and is quite possibly the highest in the universe. (Last year, we were bested by schools in Toronto and Shanghai.) The team score comprises the three top individual scores in the school, which this year were 145 (Lefkowitz), 130 (Tao), and 125 (a tie—Friedman and Yang).

In spite of the turmoil surrounding the senior team, the juniors had an excellent season. Elizabeth L. Wilmer became the first woman to correctly solve all eight individual problems at the New York State Mathematics League competition and still not win first place honors. The tiebreaker, a breathtaking arm-wrestling match lasting over three-quarters of an hour, resulted in a win for her opponent from the Southern Tier A team. Miss Wilmer has been unavailable for comment and is rumoured to be recovering in Princeton, N.J. In junior interscholastic competition, the Q team achieved first place both terms, with Captain Andy Chen, Captain Zeph Landau, and Captain Wilmer taking first, second, and third individual honors in the country.

The only member of the junior team available for comment, Mr. Landau, was found in Stuyvesant Park. Upon being asked his opinion, he gave a blank smile and replied, "I think that's really great." The one dark spot on an excellent season came when Captain Chi K. Chan quit the team, complaining of personal problems with teammates: "My name is not 'Chong'. My name is NOT 'Chong'." Distinguished semanticists are still puzzling out this remark.

It is hoped that in the coming year Stuyvesant's math team will regain the taste and style for which it has been renowned in the past.

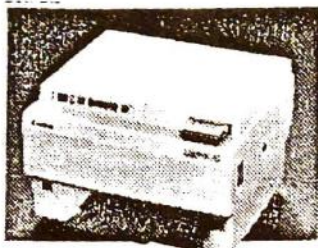
Here are some of the more interesting mathematics problems we have encountered this year. If you're the kind of person who enjoys this sort of thing, feel free to drop by room 517 at 8:00 any morning.

- (1) Find the product of all the divisors of 1,000,000.
- (2) In the expansion of $(1+x)^n$, three consecutive coefficients are in the ratio 2:15:70. Find n .
- (3) Find the smallest natural number n such that $n/2$ is a perfect square, $n/3$ is a perfect cube, and $n/5$ is a perfect fifth power.
- (4) If $\sin x + \cos x = 1/2$, find the value of $\sin^3 x + \cos^3 x$.
- (5) In triangle ABC, angle bisectors AD and BE intersect at point P. Points P, D, E, and C lie on a circle. Find ED/DP.
- (6) Do there exist 14 consecutive integers such that each is divisible by a prime p , $2 \leq p \leq 11$?
- (7) Find positive integers $a_1, a_2, \dots, a_n$ such that $a_1 + a_2 + \dots + a_n = 1000$ and the product $a_1 a_2 \dots a_n$ is as large as possible.

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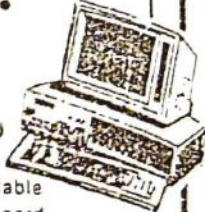
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Prime Number Distribution

by Darien G. Lefkowitz

A positive integer is called a prime number if its only divisors are itself and unity. The study of the frequency and distribution of these numbers is perhaps the oldest branch of number theory; in recent times the question of prime distributions has engendered the beautiful synthesis of analysis and number theory in the analytic theory of numbers. In this space we shall present a number of the more interesting results to be found in this discipline, without too much worry over the inclusion of formal proofs.

First, let us recall a celebrated theorem of Euclid:

Theorem: There are infinitely many primes.

proof: Suppose there were only finitely many—say $p_1, p_2, \dots, p_n$. Then the number $N = p_1 p_2 p_3 \dots p_n + 1$ is one more than a multiple of each prime, and thus has no prime factor at all. This is ridiculous.

In fact, if $\{p_1, p_2, \dots, p_n\}$ is any set of primes, then there must be another prime which does not exceed $p_1 p_2 p_3 \dots p_n + 1$. This observation can be used to obtain an estimate on $\pi(n)$ —the number of primes not exceeding n . The result is $\pi(n) \geq 1 + \log_2(\log_2 n)$.

To see this, let p_j denote the j^{th} prime. Then we have

$$p_1 = 2$$

$$p_2 \leq 2 + 1 < 2^2$$

$$p_3 \leq p_1 p_2 + 1 < 2^{1+2} + 1 < 2^{2^2}$$

$$p_4 \leq p_1 p_2 p_3 + 1 < 2^{1+2+4} + 1 < 2^{2^3}$$

and in general,

$$p_m < 2^{2^{m-1}}.$$

From this, it is not difficult to see that, if $p_m \leq n < p_{m+1}$, then

$$m = \pi(n) \leq 1 + \log_2(\log_2 n)$$

This asserts, for example, that the number of primes less than a googol is at least $1 + \log_2(\log_2 10^{100}) \approx 9$. Euclid's trick thus provides us with a very crude lower bound on $\pi(n)$.

A stronger result is the

Theorem: $\pi(n) \geq \log_2 n - 2$.

Lemma: At least $1/4$ of the integers $1, 2, \dots, n$ are square-free (i.e., have no square factor).

proof: At most $n/4$ of these numbers are multiples of four.

At most $n/9$ are multiples of nine.

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At most n/r^2 are multiples of r^2 .

The total number of integers $\leq n$ with square factors is thus at most

$$\begin{aligned} n(1/4 + 1/9 + 1/16 + \dots) &\leq n(1/4 + 1/2 \cdot 3 + 1/3 \cdot 4 + 1/4 \cdot 5 + \dots) \\ &= n[1/4 + (1/2 - 1/3) + (1/3 - 1/4) + (1/4 - 1/5) + \dots] \\ &= (3/4)n \end{aligned}$$

Therefore, at least $1/4$ of these numbers have no repeated factor.

proof of theorem: Let $m = \pi(n)$, and suppose the primes $\leq n$ are $p_1, p_2, \dots, p_m$.

Then, by regarding a square free number less than n as the product of a subset of $\{p_1, p_2, \dots, p_m\}$, we see from the lemma that this set has at least $n/4$ subsets. That is,

$$2^m \geq n/4,$$

whence

$$m = \pi(n) \geq \log_2(n/4) = \log_2 n - 2.$$

Q.E.D.

How can we obtain something closer to the true value of $\pi(n)$?

To do this, we shall embark upon a temporary digression—consider the following question:

What is the exponent of p ($p \leq n$) in the prime factorization of $n!$? This is a standard “math team problem”, and its answer is

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$$f_p(n) = \sum_{k=1}^{\infty} [n/p^k] \leq \sum_{k=1}^{\infty} n/p^k = n(p^{-1} + p^{-2} + p^{-3} + \dots) = \frac{n}{p-1}.$$

Note also that $f_p(n) \geq [n/p] \geq (n/p) - 1$. Since we are concerned with large values of n (and p), and have seen that $(n/p) - 1 \leq f_p(n) \leq n/(p-1)$, we may as well write $f_p(n) \approx (n/p)$, whereupon

$$n! = 2^{f_2(n)} 3^{f_3(n)} \dots p^{f_p(n)} \quad (p \text{ the biggest prime } \leq n) \\ \approx 2^{(n/2)} 3^{(n/3)} \dots p^{(n/p)}.$$

Now, $\log n! \approx n \sum_{p \leq n} \frac{\log p}{p}$. But $\log n! = \sum_{k=1}^n \log k$ can also be approximated by a definite integral; to wit,

$$\sum_{k=1}^n \log k \approx \int_1^n \log x \, dx = [x(\log x - 1)]_1^n = n \log n - n + 1. \text{ Again, for}$$

huge values of n , the number $n-1$ is absurdly small when compared with $n \log n$, and we write $n \log n - n + 1 \approx n \log n$. The transitivity of $\approx$ now informs us that

$$n \log n \approx n \sum_{p \leq n} \frac{\log p}{p}, \text{ or}$$

$$\log n \approx \sum_{p \leq n} \frac{\log p}{p}.$$

But wait! Now $\log n = \int_1^n dx/x \approx \sum_{k=1}^n 1/k$, and finally

$$\sum_{p \leq n} \frac{\log p}{p} \approx \sum_{k=1}^n 1/k.$$

The point of all this is that, if the last expression is written as

$$\sum_{k=1}^n \frac{\log k}{k} \cdot \frac{1}{\log k}, \text{ the factor } (1/\log k) \text{ can be interpreted as the}$$

the "probability" that k is prime, or, if you prefer, the "average number of primes per unit interval" at k . Gauss made the same observation (from empirical data) around 1800, and inferred, as any good calculus student would, that

$$\pi(x) \approx \int_2^x \frac{dt}{\log t}.$$

This result was actually proven mathematically in 1896, (independently by Hadamard and de La Vallée Poussin), and has become known as the prime number theorem. It is usually stated in the equivalent form $\pi(x) \approx x/\log x$.

Here and above, the symbol " $\approx$ " becomes meaningful only when interpreted in the sense of asymptotic equivalence (usually denoted by " $\sim$ "). For example, the formula $\pi(x) \approx x/\log x$ means that $\frac{\pi(x)}{x/\log x}$ tends to unity as x tends to infinity.

The proof that the two forms of the prime number theorem are equivalent is a bit too complicated to include here.

We have seen that there are infinitely many prime numbers; a measure of strength of this infinitude is indicated by the fact that the sum of the

reciprocals of the primes $(\sum_p 1/p)$ diverges. The usual proof of this

involves the Riemann zeta-function, defined by $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$. The

connection of the ζ -function with the primes numbers is established by the infinite product formula

$$\begin{aligned}\zeta(s) &= (1+2^{-s}+2^{-2s}+2^{-3s}+\cdots)(1+3^{-s}+3^{-2s}+\cdots)(1+5^{-s}+5^{-2s}+\cdots)\cdots \\ &= \prod_p (1-p^{-s})^{-1}, \text{ where } p \text{ is to run through all the primes.}\end{aligned}$$

From this formula and Taylor's theorem, one can obtain

$$\log \zeta(s) = \sum_{m,p} m^{-1} p^{-ms} \quad (\text{here } m \text{ is supposed to run through the positive integers, and } p \text{ through the primes.})$$

$$\text{and then } -\frac{\zeta'(s)}{\zeta(s)} = \sum_{m,p} (\log p)/p^{ms}.$$

These formulæ suggest that the points where $\zeta(s)=0$ are in some way relevant to the distribution of the primes. Indeed, it can be said that the problem of the distribution of the primes is equivalent to the location of the zeros of the zeta function.

A generalization of Riemann's zeta function was discovered by Dirichlet, and is known as the Dirichlet L-series. He used these series to isolate the primes in a particular arithmetic sequence, and was able to prove that the sum of the reciprocals of the primes of the form $p = ak+d$ will diverge provided that a and d are relatively prime. In particular, then, any arithmetic sequence with common difference relatively prime to the first term contains infinitely many primes (Dirichlet's Theorem). This important theorem concerning the regularity of the distribution of the primes is an example of a theorem for which there is no known proof which does not employ analytic techniques. However, many special cases can be proved by simple extensions of Euclid's trick. As an example, we prove that infinitely many primes are of the form $4k+3$. For suppose there were only finitely many — say $p_1, p_2, \dots, p_n$, and let $N = 4p_1 p_2 \cdots p_n - 1$. Then N has no prime factor of the form $4k+3$. But this contradicts the fact that N is 3 more than a multiple of 4, and so there must have been infinitely many such primes. The interested reader may wish to use this sort of manipulation to prove that there are also infinitely many primes of the form $4k+1$.

A Note on the Text

We are indebted to the Apple Computer Corporation for their generous donation of the Macintosh computers on which this magazine was produced. This magazine was printed with an Apple typeset quality LaserWriter printer on the premises of Stuyvesant High School. The text was set in the Times typeface; "Times" is a trademark, valid under applicable law, of the Allied Corporation and is used by Apple under license from Mergenthaler Linotype. The use of this trademark by the Mathematics Department of Stuyvesant High School does not entail any right, title, or interest in or to this trademark. Apple, Macintosh, and LaserWriter are trademarks of Apple Computer, Inc.

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1. Suppose n cars drive into town and intend to park on the same block. The block contains exactly n parking spaces in a single file, so of course it will be easy for all the cars to park. However, each of the n drivers has a first choice of a parking spot without knowing the other driver's choices. If, when a driver arrives at his chosen spot, it is already occupied he moves on to the next unoccupied spot, if available, to park. What is the probability that all drivers park successfully? Note: This is all on a one-way street, so the drivers may not back up to an unoccupied spot.
2. We play the following game. Initially n poker chips are placed at the 0 point on the number line. Next, if n is even, we divide the pile in half and place half the pile at -1 and the other half at $+1$. On the other hand, if n is odd, we leave one chip alone and split the remaining $n-1$ chips in half as before. We continue in the same way with the resulting piles. For example, if $n=6$, we have the following pattern of chips

6
303
11211
12021
20202
1020201
1102011
1110111,

after which there is obviously no further change.

- (a) Show that the game terminates in a finite number of moves.

Let $f(n)$ denote the number of moves in the game. By part (a), $f(n)$ is finite. Show that

(b) $f(2n) = f(2n+1)$.

(c) $f(n)/n^2$ is bounded (i.e., $f(n)/n^2 \leq c$ for some number c).

Send your solutions to: MATH SURVEY CONTEST c/o Lori Smith
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